

# XUEQING GAO

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## Education

**Tongji University** (*Global Rank: 124, U.S. News*)

Shanghai, China

Major GPA – 90.4/100.0 (Rank: 3/46), Overall GPA – 87.2/100.0

Previously in Mechatronic Engineering, Chinesisch-Deutsche Hochschule für Angewandte Wissenschaften Sep. 2021 – Jul. 2022

B.E. in Optoelectronic Information Science and Engineering, School of Physics Science and Engineering Sep. 2022 – Jul. 2025

## Work Experience

**Deep Learning for Diffusion MRI Microstructure Estimation**

July 2023 – Feb. 2025

*Research Intern — Mentor: Prof. Yuchuan Qiao and Prof. Yonggang Shi*

*Fudan University & USC (CA, USA)*

- Developed physics-based deep learning methods (UFO-3) for reconstructing fiber Orientation Distributions Function (fODF) from highly limited diffusion MRI data.
- Combined biophysical modeling, spherical harmonics, and optimization-based constraints to improve interpretability, accuracy, and generalizability across populations and acquisition protocols.
- Achieved sub-second inference and strong performance on Chinese-HCP, and simulated datasets, outperforming conventional model-based and recent learning-based methods.

**Transmission-Aware Unpaired Test-Time Haze Refinement**

Sep. 2025 – Apr. 2026

*Full-Time Research Assistant — Mentor: Prof. Haibin Ling*

*Stony Brook Univ. & Westlake Univ.*

- Proposed an unpaired, physics-aware dehazing deep learning framework that jointly estimates the scattering coefficient  $\beta$  and clean radiance via Schrödinger Bridge transport under Atmospheric Scattering Model constraints.
- Developed a differentiable fog-density evaluator and test-time adaptation strategy that refines  $\beta$  and the restored image per instance using physics-guided reconstruction losses without updating network weights.
- Achieved state-of-the-art no-reference performance among unsupervised methods on real-world benchmarks, improving perceptual quality and structural fidelity under complex haze conditions.

## Selected Publications

- [Oral Presentation, Top 2.2%] “UFO-3: Unsupervised three-compartment learning for rapid fiber orientation distribution function estimation from diffusion MRI,” **Xueqing Gao**<sup>†</sup>, Rizhong Lin<sup>†</sup>, Jianhui Feng, Yonggang Shi, Yuchuan Qiao. Accepted to the 28th *International Conference on Medical Image Computing and Computer Assisted Intervention* (MICCAI 2025), a top-tier conference in medical AI.
- “Scattering-Aware Unpaired Test-Time Haze Refinement,” **Xueqing Gao**, Zhijian Wu, Bingyao Huang, Yefeng Zheng (*IEEE Fellow*), Haibin Ling (*IEEE Fellow*). Planned to submit to AAAI 2027.
- “Empowering Biomedical Research with Foundation Models in Computational Microscopy: A Systematic Review,” Di Ding<sup>†</sup>, Tianliang Yao<sup>†</sup>, Genyi Feng, **Xueqing Gao**, Rong Luo, Xusen Sun. *Advanced Intelligent Systems* (IF: 6.8), accepted for publication.

## Selected Project Experience

**OpenPixel-RPG — AI-Driven Pixel-Art Open-World JRPG** Personal Project — 170+ stars, 15 forks 📄

- Built three-mode AI game generation: (1) photo upload → pixel JRPG map; (2) NYC landmark → pre-generated tiles; (3) any global address → OSM Overpass + satellite + Google 3D Tiles + AI style transfer → playable world.
- Engineered OSM whitebox pipeline (Python): Overpass API → isometric geometry render (roof/wall shading) → geometric blueprint constraining Gemini pixel-art generation, preserving real-world building layouts.

- Implemented multi-model Gemini orchestration: 2.5 Pro (world design), 3.1 Flash (pixel map + sprites), 3.1 Pro (vision review), 2.5 Flash (runtime NPC decisions + dialogue + memory).
- Built full-stack backend with Node.js + Express + SQLite + WebSocket, React 19 + Phaser 3 frontend; Phaser handles isometric rendering, A\* pathfinding, and camera following.

## Harrier-SwissLaw-Retrieval — Cross-Lingual Legal Retrieval Agent System Kaggle Competition

- **Task:** Kaggle Omnilex Agentic Legal Retrieval — English legal queries retrieve precise citation strings from 170K German statutes and 2.4M multilingual case laws, evaluated by Macro F1 under 12-hour offline constraint.
- **Embedding domain adaptation:** Fine-tuned Microsoft Harrier-OSS-v1 (27B, SOTA on multilingual MTEB v2) with 4-bit NF4 quantization + LoRA (rank=64, 454M trainable params) using PyTorch DDP on 6×RTX 5090 32GB — Stage 1 warm-up with in-batch negatives, Stage 2 hard negative mining.
- **Hard negative mining:** Reproduced NV-Retriever TopK-PercPos strategy — TF-IDF top-50 candidates, retaining those below 95% of positive score as hard negatives (5 per query), processed in seconds via scikit-learn sparse matrices on 3570 training pairs.
- **Hybrid retrieval & reranking:** Dense (fine-tuned Harrier-27B + FAISS, top-100) + Sparse (TF-IDF for statutes and cases, top-100) → Reciprocal Rank Fusion (RRF) to top-60 candidates → full fine-tuned BGE-Reranker-v2-m3 (570M Cross-Encoder) with score-gap dynamic truncation for adaptive output.

## Professional Activities & Community Engagement

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**Academic Service:** Reviewer, NeurIPS 2026.

### Open-Source Contributions:

**Awesome Computational Imaging** [Link]

*Author and Maintainer*

Curated a learning and implementation series on modern computational imaging, covering implicit neural representations (SIREN, NeRF, FFN), Gaussian Splatting, diffusion models for inverse problems, and plug-and-play methods.

## Awards

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- **Zhuiyuan Scholarship** (awarded for excellence in optical research; funded by Prof. Zhanshan Wang), 2024.
- **Outstanding Undergraduate Scholarship, Tongji University** (awarded to undergraduates based on academic performance), 2022–2024.

## Skills

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- **Vibe Coding:** Claude Code, Cursor.
- **Programming Languages:** Python, C/C++, MATLAB, Julia.
- **Languages:** Chinese, English, German

## Talks

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- **[2025-09]** “UFO-3: Unsupervised Prediction of White Matter Fibers”  
Medical Image Computing Seminar (MICS), [online](#).
- **[2025-09]** “UFO-3: Unsupervised Prediction of White Matter Fibers”  
MICCAI 2025, Daejeon, South Korea, [\[Slides\]](#).

Last updated: June 26, 2026.